Session 6: Protein Structure

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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An α-helix is held together mainly by …

A Disulphide bonds

B Backbone hydrogen bonds, residue i to i + 4

C Salt bridges between side chains

D Hydrogen bonds between neighbouring strands

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What does a protein's contact map show?

A Which residues are close to each other in 3D

B Which proteins interact in the cell

C The hydrophobicity along the sequence

D Where in the cell the protein is found

Answer now

Scan the code, or go to

quiz.bioinfo.se/live

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During evolution, which usually changes more slowly?

A The sequence

B Both change equally fast

C The structure

D Neither changes

Answer now

Scan the code, or go to

quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Four levels

sequence to 3D, contact maps

Comparing in 3D

RMSD, TM-score, DALI, Foldseek

Domains and states

CATH, T and R, prediction

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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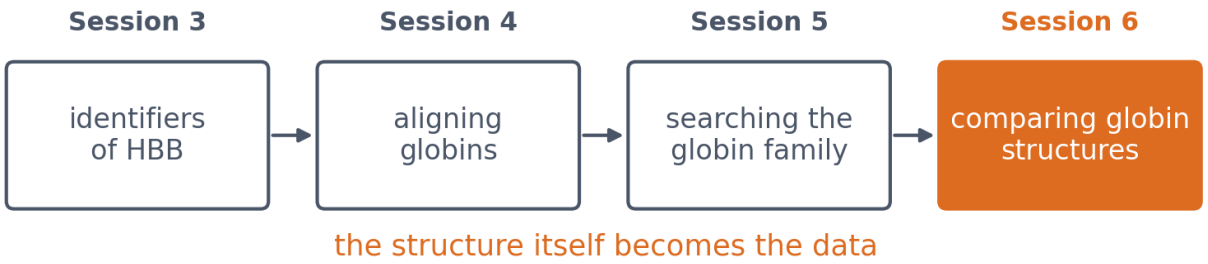
What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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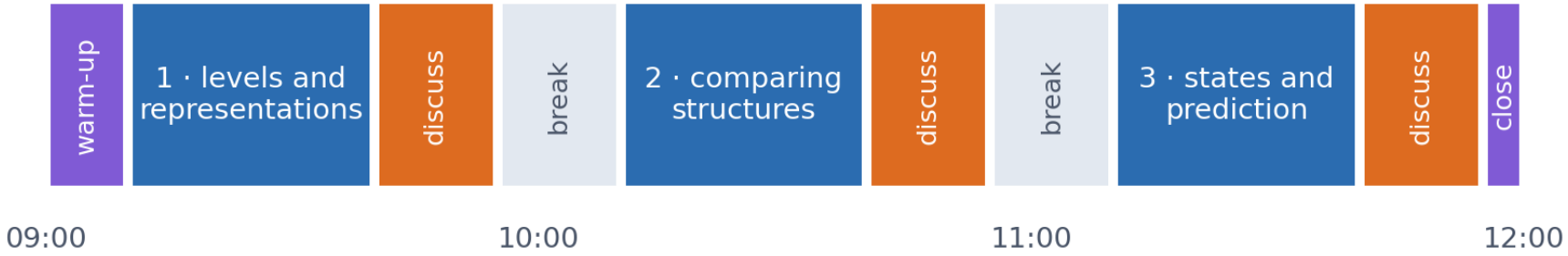
What does a sequence encode in 3D?



- **Today the structure is the data**
- Coordinates → distances, domains, interfaces
- Sessions 7-12: can a computer predict it?

See course.bioinfo.se/session06.html, 'Learning goals'.

Three blocks: about 30 min talk and 15 min discussion



Block 1: the four levels of structure and how to represent a structure. Block 2: comparing structures (alignment, RMSD, TM-score, DALI, Foldseek). Block 3: domains, conformational states, and structure as a prediction target.

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

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BLOCK 1 · ABOUT 30 MIN

Sequence becomes structure

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Four levels of structure, all in haemoglobin

primary

the sequence

VHLTPEEKSA
VTALWGKVNV
DEVGGEALGR
LLVVYPWTQR
FFESFGDLST
PDAVMGNPKV
…

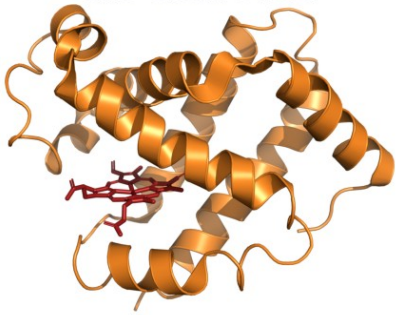
secondary

one α-helix



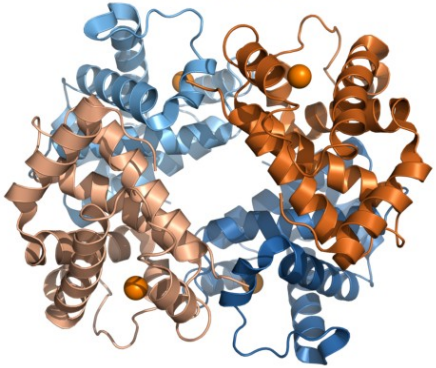
tertiary

one chain's fold



quaternary

four chains



Primary: the amino-acid sequence. Secondary: local backbone shapes (α-helix, β-strand, coil). Tertiary: the 3D fold of one chain. Quaternary: several chains packed together. Rendered from PDB 2HHB with PyMOL.

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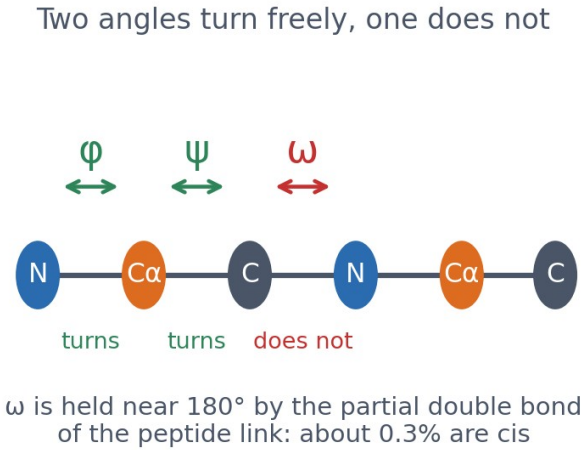
Each level comes with its own computational task

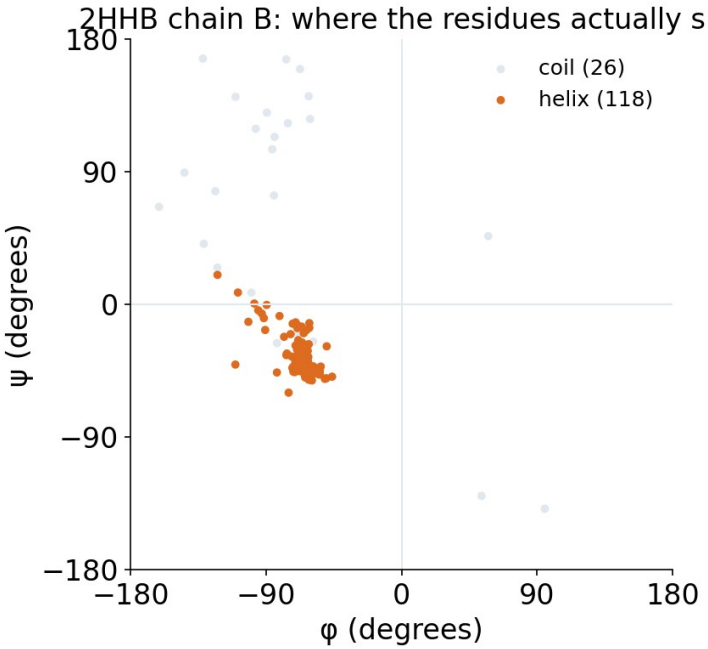
level	what it is	computational task	where
primary	sequence	alignment, BLAST, profiles	Sessions 3-5
secondary	helix, strand, coil	prediction from sequence	Sessions 7-11
tertiary	one chain's fold	comparison, prediction	today; Session 12
quaternary	several chains	interfaces, complexes	today; Session 12

The course follows the levels in order: sequences (Sessions 3-5), secondary structure (Sessions 7-11), folds and complexes (today and Session 12).

See course.bioinfo.se/session06.html, Table 6.1.

Flexible means two angles, not three





φ turns about N–Cα, ψ about Cα–C. ω, about the peptide bond itself, barely turns at all. Right: every residue of 2HHB chain B at its measured φ and ψ, coloured by the secondary structure PyMOL assigns it.

Angles read from 2HHB with PyMOL; drawn by tools/builders/make_session06_slide_figures.py.

Helix, strand, coil

Ubiquitin (PDB 1UBQ)



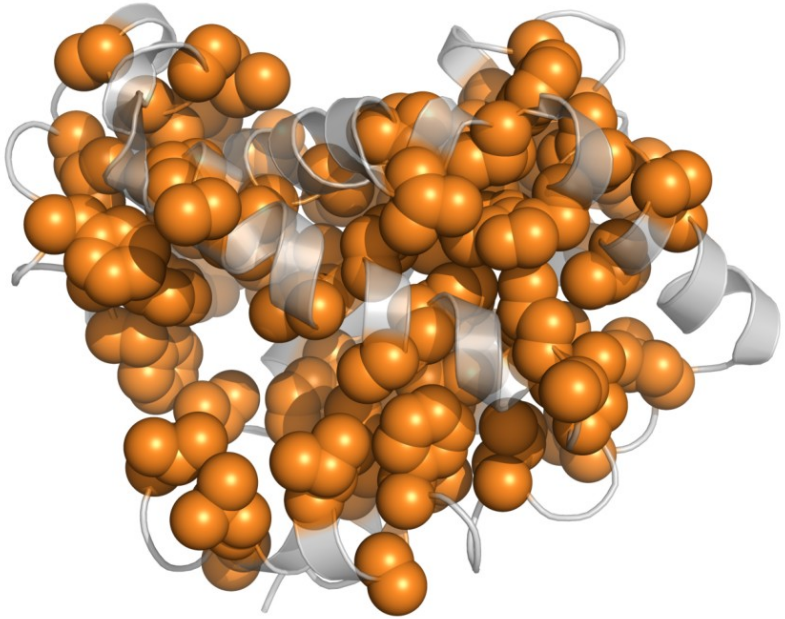
helix strand coil

- **Helix: H-bonds between residues i and i+4**
- **Strand: H-bonded edge to edge in a sheet**
- Coil: everything else
- DSSP assigns them from coordinates

Ubiquitin (PDB 1UBQ), rendered with PyMOL.

The fold hides its hydrophobic side chains

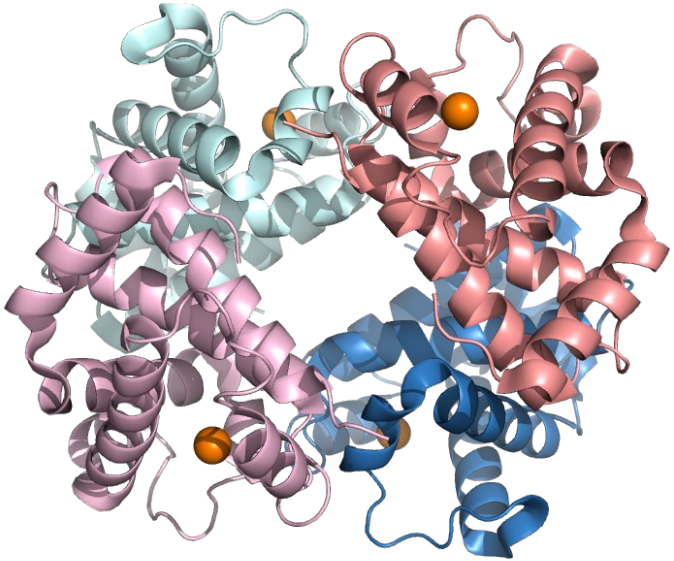
β-globin: hydrophobic side chains (orange)



- **Hydrophobic effect: non-polar side chains away from water**
- Plus H-bonds, electrostatics, van der Waals
- Many proteins work in complexes

β-globin (2HHB chain B); Ala, Val, Ile, Leu, Met, Phe, Trp, Cys as spheres.

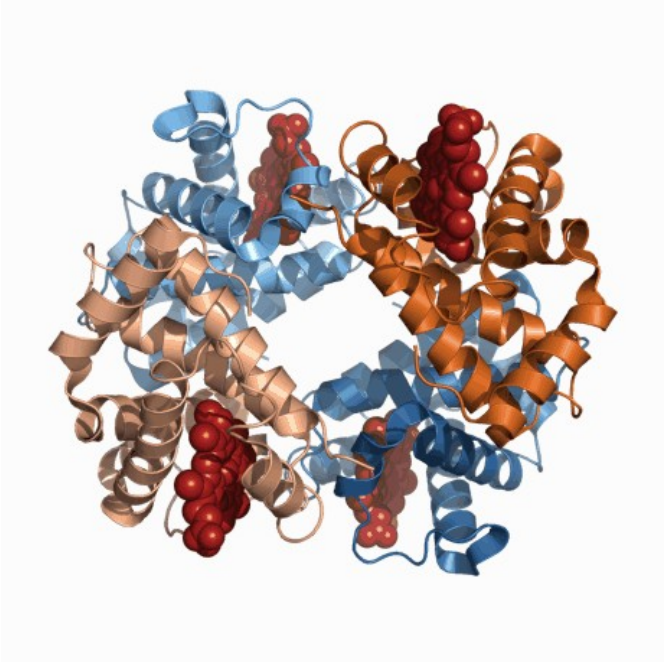
Haemoglobin: two α and two β chains



- **α: 141 residues; β (HBB): 146**
- One haem per chain (orange: iron)
- **One β chain: 80.8% helix, 0% strand**

PDB 2HHB, rendered with PyMOL; numbers from session06-protein-structure-in-code.ipynb.

Haemoglobin, turning



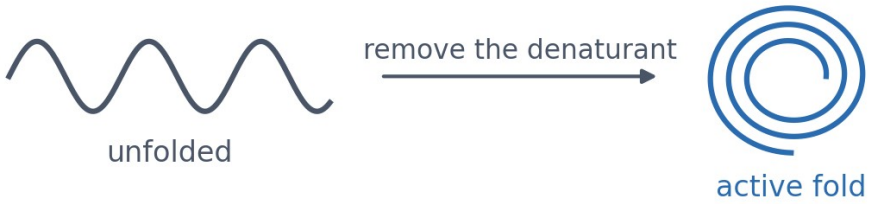
Two α chains (blue) and two β chains (tan), with the four haem groups as red spheres. One revolution, ten degrees a frame.

2HHB, drawn by tools/builders/make_session06_slide_figures.py. Animated in PowerPoint and LibreOffice; the PDF keeps the first frame.

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Does the sequence determine the structure?

Anfinsen, 1960s: ribonuclease refolds on its own



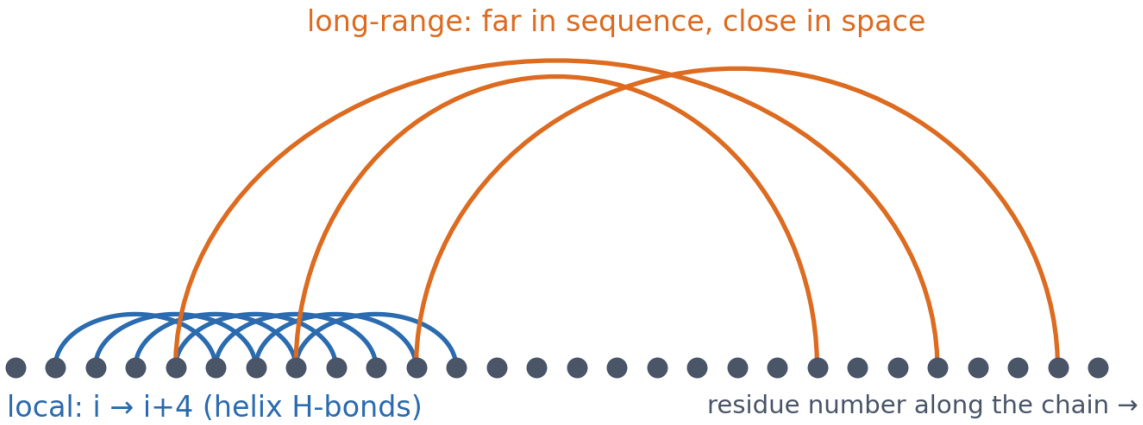
- **Often yes**
- That makes prediction possible
- Not a law for every protein

But not always: chaperones · ligands and partners · alternative states · disorder

See course.bioinfo.se/session06.html, 'Does the sequence determine the structure?'.

SciLifeLab

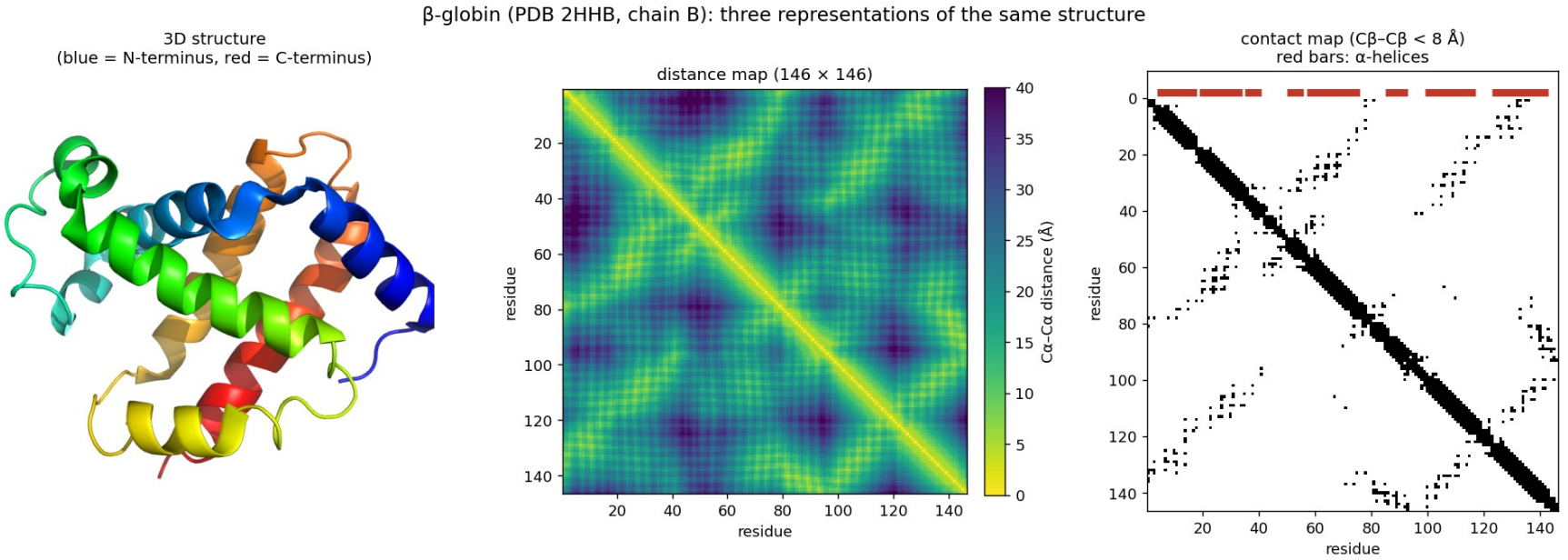
Local is easy, long-range is hard



- **Local: secondary structure**
- **Long-range: the tertiary fold**
- A window of sequence sees only local

See course.bioinfo.se/session06.html, 'Does the sequence determine the structure?'.

A structure as a matrix: distance and contact maps

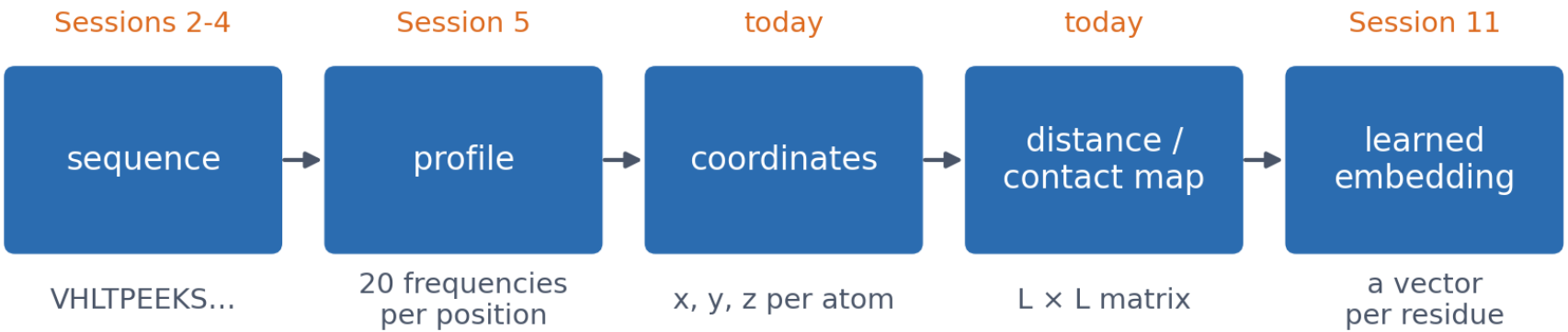


β-globin as a structure, a distance map and a contact map (Cβ within 8 Å). Diagonal band: each helix touching itself (local). Off-diagonal stripes: helices packed together (long-range). The maps do not change when the protein is rotated.

tools/builders/make_session06_figures.py; notebook.

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One protein, many representations

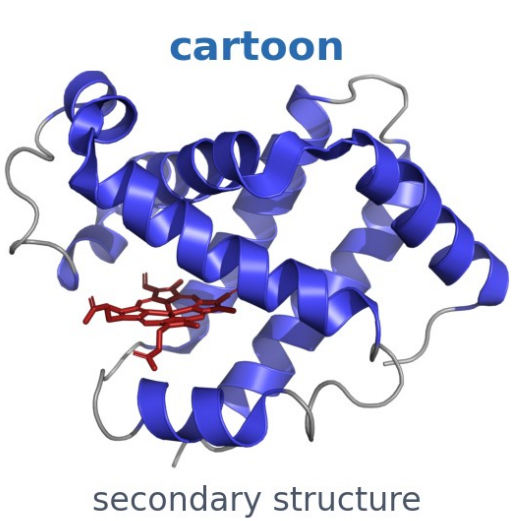


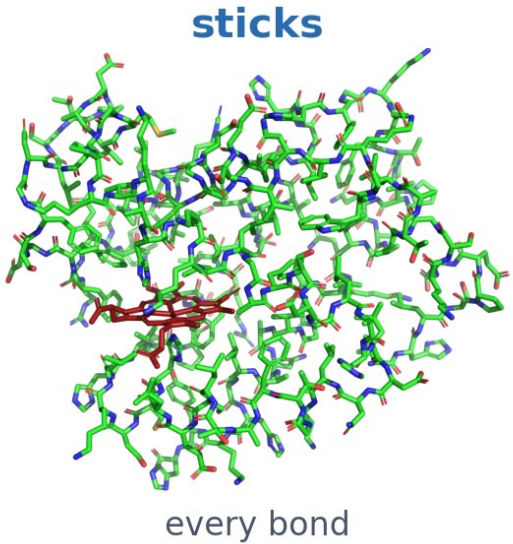
Each representation makes different patterns easy to see. Later in the course it also decides which patterns a model can easily learn.

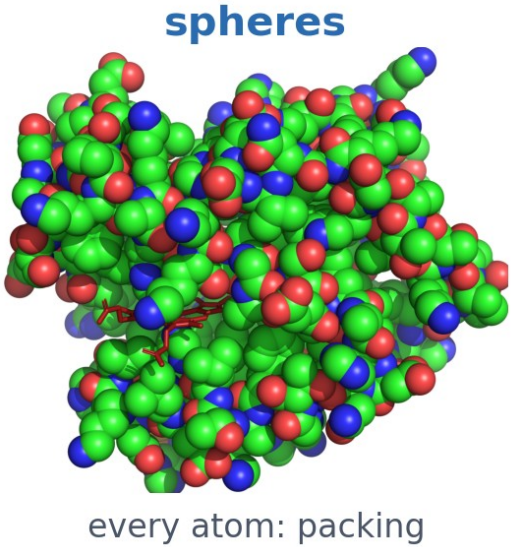
See course.bioinfo.se/session06.html, callout 'Representations'.

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PyMOL: three ways to draw the same chain





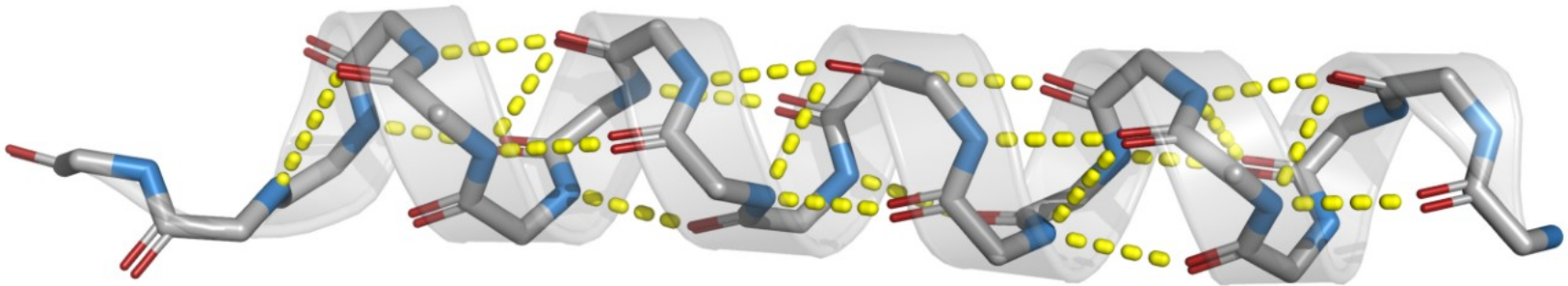


Cartoon shows the secondary structure, sticks every bond (for a side chain or a ligand), spheres every atom (for packing). The lab uses PyMOL hands-on; the notebook drives it from Python.

β-globin (2HHB chain B), haem in red.

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Hydrogen bonds: measured, not drawn



The α-helix is held by N–H···O=C bonds from residue i to residue i+4. PyMOL finds them from the coordinates with a 3.5 Å cutoff; every dash here is a real distance in 2HHB.

2HHB chain B, residues 125–144 (the H helix). Drawn by tools/builders/make_session06_slide_figures.py.

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The whole figure is eight lines

```
fetch 2hhb, async=0
create helix, 2hhb and chain B and resi 125-144
show cartoon, helix
show sticks, helix and name N+CA+C+O

# N-H to O=C, i to i+4, within 3.5 Angstrom
dist hb, helix and name N, helix and name O, 3.5, mode=2
set dash_color, yellow
```

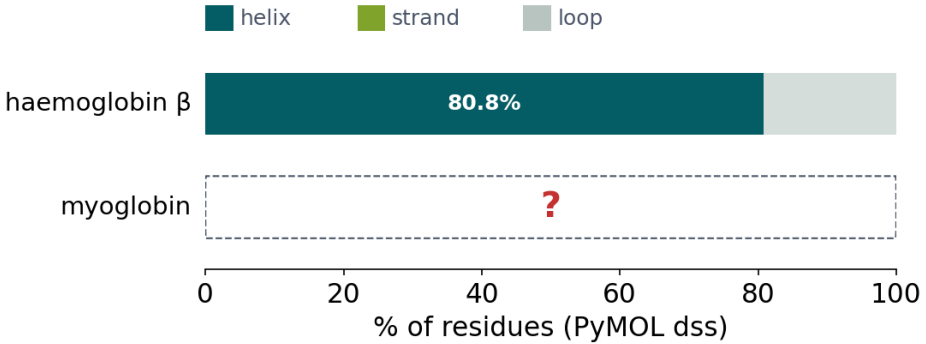
The same commands the figure builder runs; `pymol -cq script.pml` reproduces the picture.

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What does dss find for myoglobin?

Haemoglobin β: 80.8% helix, 0% strand, but it works only as four chains. Myoglobin: the same fold, one chain.





A About 80% helix, 0% strand: the same

B About 50% helix: some helices unfold

C About 40% helix, 30% strand

D Hardly any helix or strand: mostly loop

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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What does dss find for myoglobin?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

What does dss find for myoglobin?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-1.ipynb

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What does dss find for myoglobin?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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What does dss find for myoglobin?

Results, and who explains

Both rounds side by side.

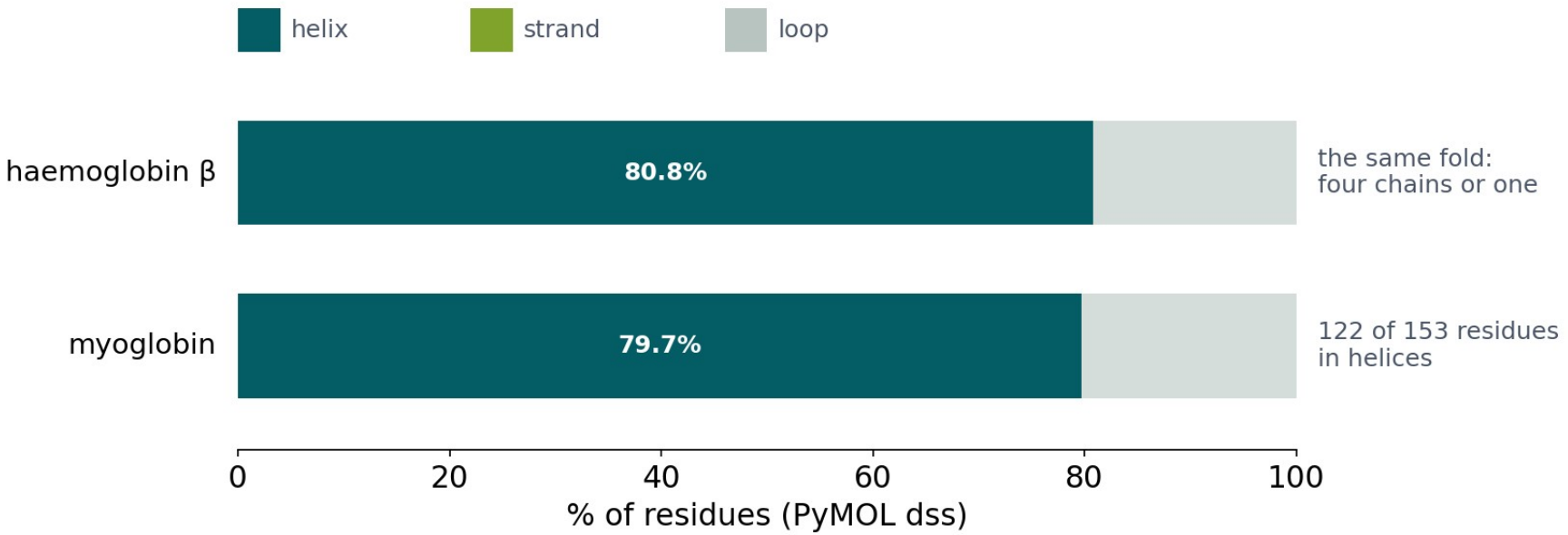
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-1.ipynb

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The answer: the same fold, alone or in a tetramer



Myoglobin is 79.7% helix and 0% strand, haemoglobin's β chain 80.8% and 0%. Secondary and tertiary structure are properties of one folded chain; quaternary structure is a separate level on top. Haemoglobin needs four chains for cooperative oxygen binding, not for its fold. And a result of exactly zero is worth a second look.

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BLOCK 1 DONE

Break

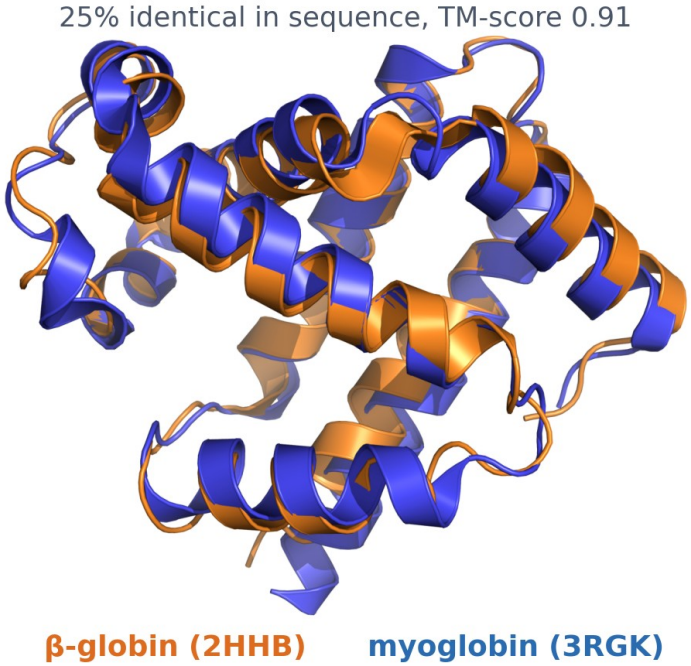
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BLOCK 2 • ABOUT 30 MIN

Comparing proteins in 3D

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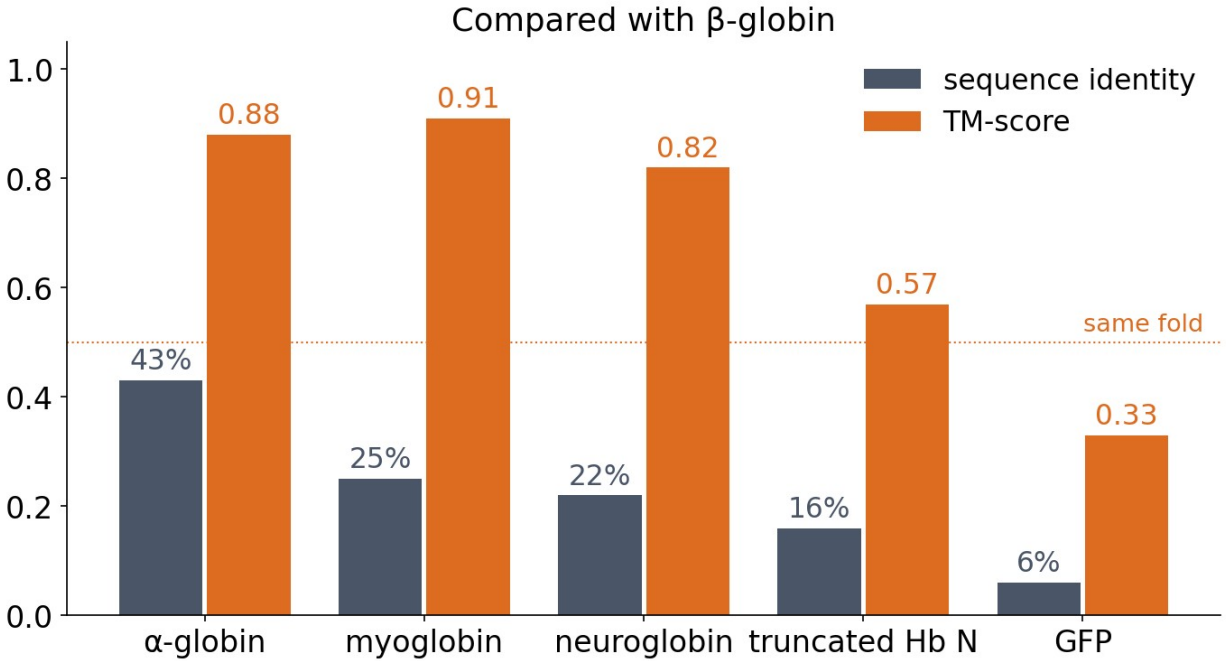
A structure alignment pairs residues by their place in 3D



- **Sequence alignment: from substitution scores**
- **Structure alignment: from coordinates**
- Both are hypotheses about correspondence

Superposition with PyMOL; TM-score from the notebook (TM-align, tmtools).

Sequence identity fades, the fold stays

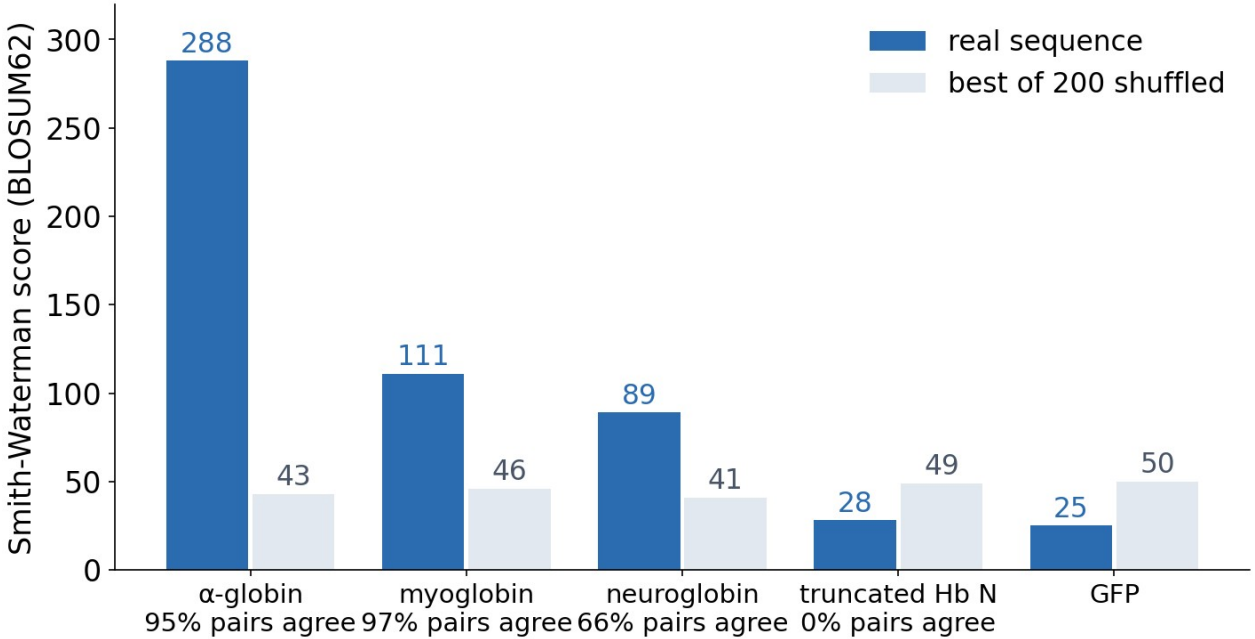


- **Five proteins vs β-globin**
- Identity 43% → 6%
- TM-score stays above 0.5 for all globins

Notebook: Smith-Waterman/BLOSUM62 and TM-align (tmtools). TM-scores normalised by the length of β-globin.

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When sequence signal is gone, structure still finds the relative

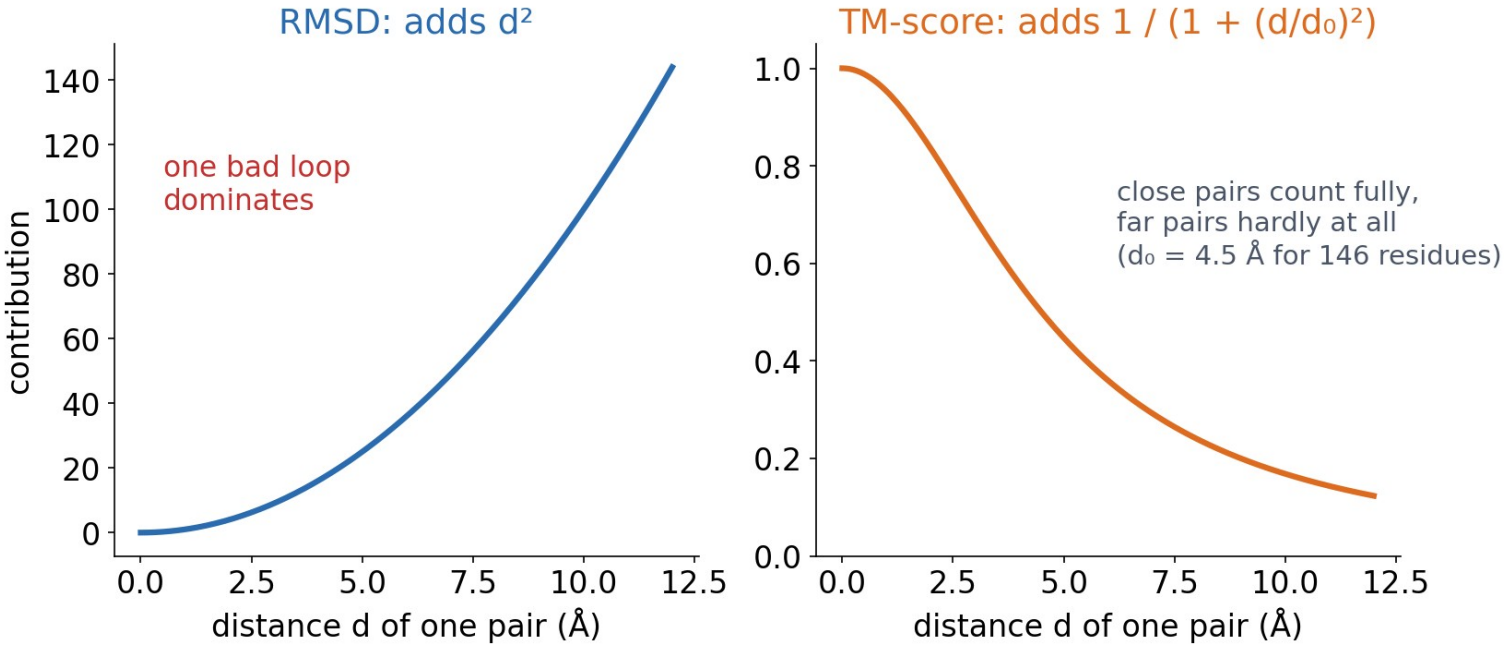


- **Truncated Hb: no better than shuffled**
- 0% of its pairs right
- **Structure is more conserved than sequence**

Notebook: best Smith-Waterman score against the best of 200 alignments to shuffled sequences.

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RMSD and TM-score weight a bad pair differently

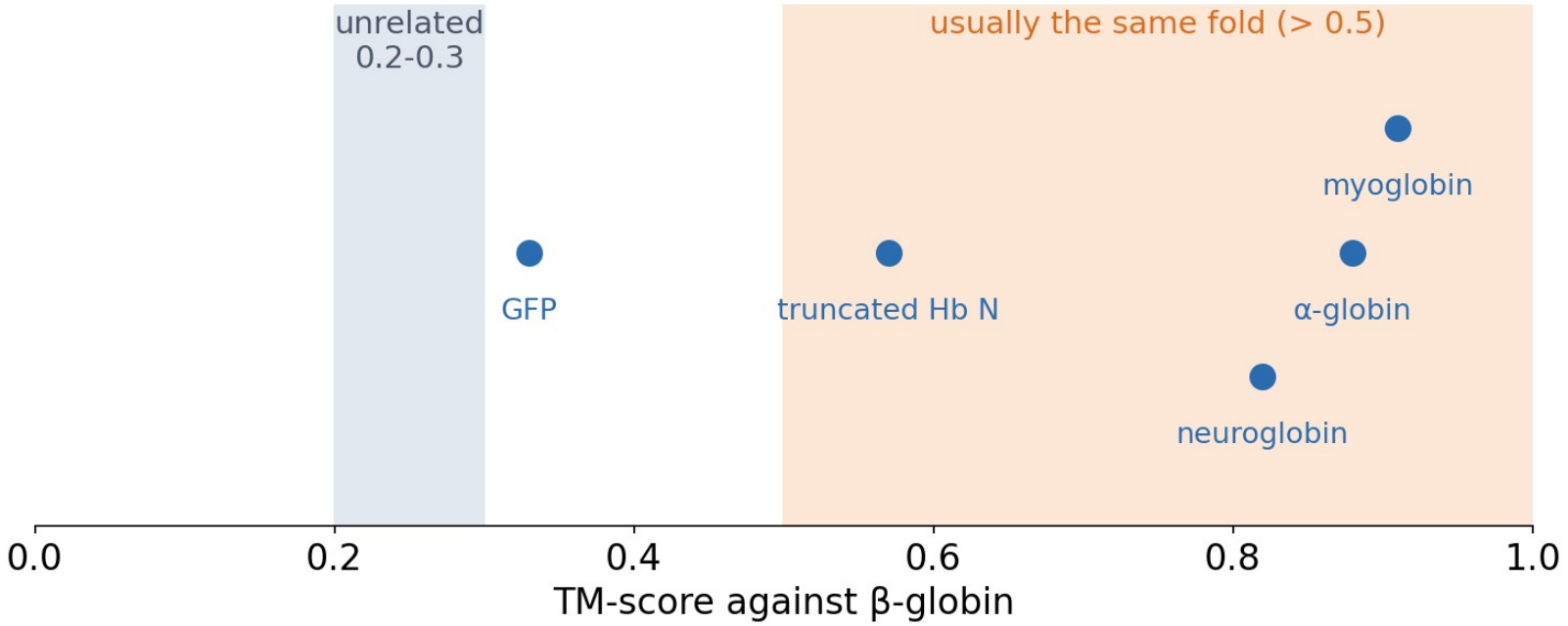


RMSD: the root-mean-square distance between corresponding atoms after superposition; it grows with length, one misplaced loop inflates it, and it means little without the number of aligned residues. TM-score: weights each pair by how close it is and normalises by length, from 0 to 1.

See course.bioinfo.se/session06.html, 'Measuring structural similarity: RMSD and TM-score'.

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Reading a TM-score

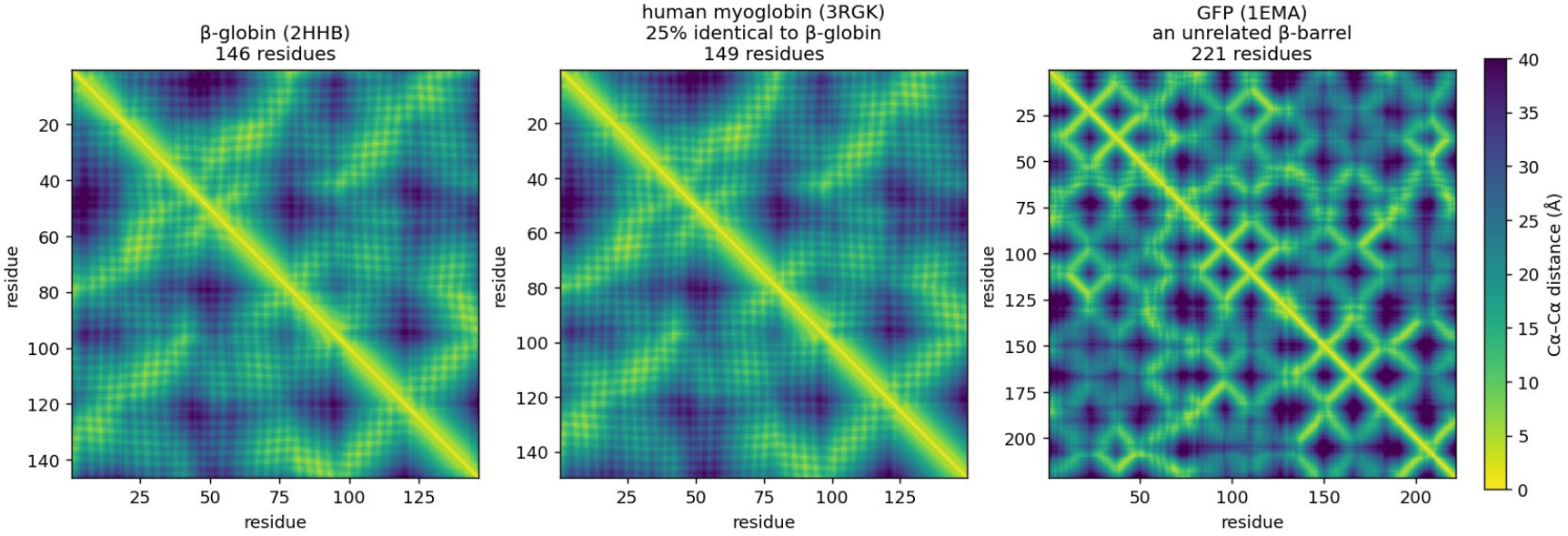


Unrelated proteins typically score 0.2-0.3; above about 0.5 usually means the same fold. A rule of thumb, not a law: scores in between need other evidence.

TM-scores from the notebook.

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DALI: comparing patterns of internal distances



β-globin and myoglobin (25% identical) give almost the same distance map; GFP's β-barrel does not. Sequence alignment compares residues; DALI compares residue-residue geometry.

Holm & Sander (1993). Figure: tools/builders/make_session06_figures.py.

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Searching by sequence, searching by structure

Sequence (Session 5)

BLAST, MMseqs2

- Search with the amino-acid sequence
- Billions of sequences

Structure (today)

Foldseek

- Search with a 'structural sequence'
- Hundreds of millions of structures in minutes

van Kempen, Steinegger et al. (2024), Nature Biotechnology 42:243-246.

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Foldseek turns a structure into a string



the 20 states were learned by a variational autoencoder, not designed by hand

Each residue's local 3D environment becomes one of 20 '3Di' letters. The structure is then searched with fast sequence-alignment machinery instead of expensive superposition.

See course.bioinfo.se/session06.html, 'Foldseek' and the callout 'Machine-learning preview'.

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A worked comparison with Foldseek

- **Same protein, two states: T (2HHB) vs R (1HHO)**
- α vs β globin (43% identical): TM 0.90-0.91
- **2HHB vs GFP: no hits, even with a loose threshold**

0.98
TM-score
T vs R

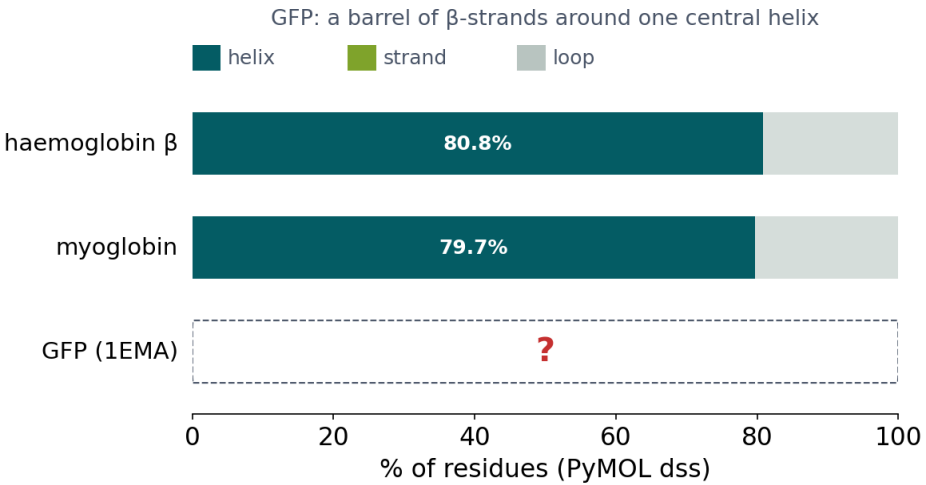
0.7 Å
RMSD
T vs R

Foldseek run locally on 2HHB, 1HHO and 1EMA (course.bioinfo.se/session06.html).

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What does dss find for GFP?

The globins: about 80% helix, 0% strand. GFP (1EMA): a barrel of β-strands around one central helix.



A About 80% helix, 0% strand, like the globins

B About 50% helix, 50% strand

C About 10% helix, 54% strand

D About 0% helix, 100% strand

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

SciLifeLab

What does dss find for GFP?

How did the room answer?

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SciLifeLab

What does dss find for GFP?

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The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-2.ipynb

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What does dss find for GFP?

Answer again

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quiz.bioinfo.se/live

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What does dss find for GFP?

Results, and who explains

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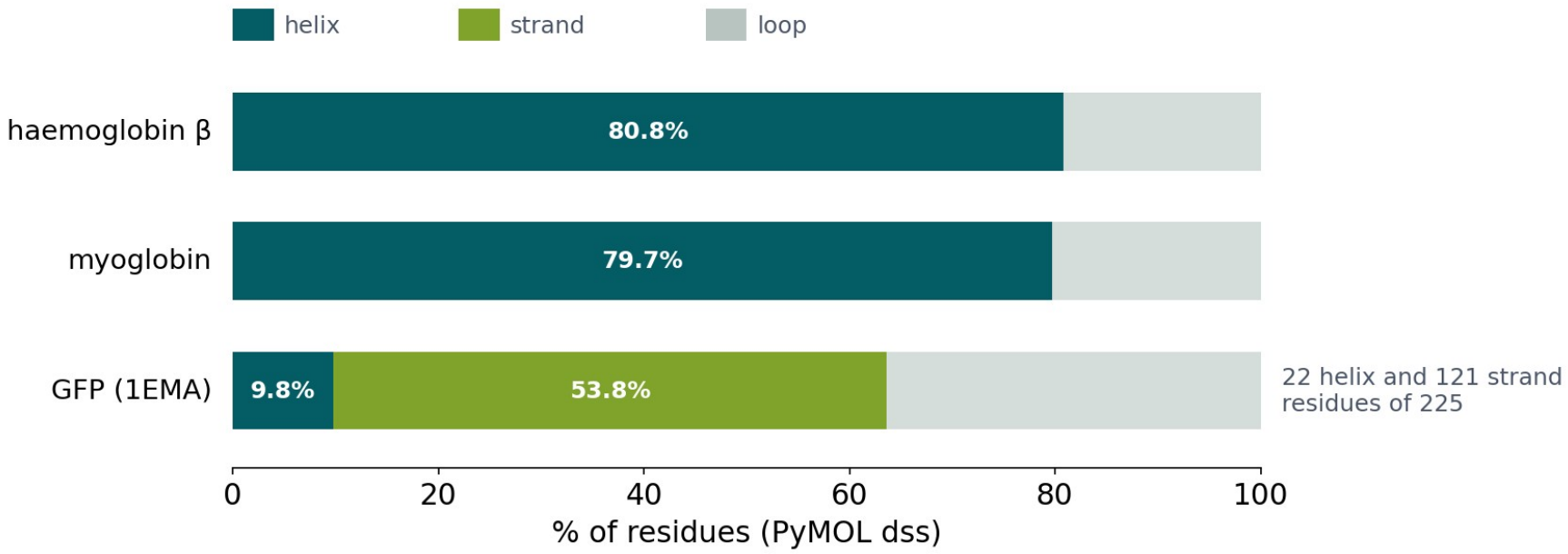
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-2.ipynb

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The answer: GFP is mostly strand



GFP is 9.8% helix and 53.8% strand, almost the mirror image of the globins. The globins are an all-α family, not a typical protein, so secondary-structure prediction cannot rely on one fixed rule: it has to learn position-specific patterns (Session 7). A copied block of code that still names the old object is a common mistake; a function called twice avoids it.

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BLOCK 2 DONE

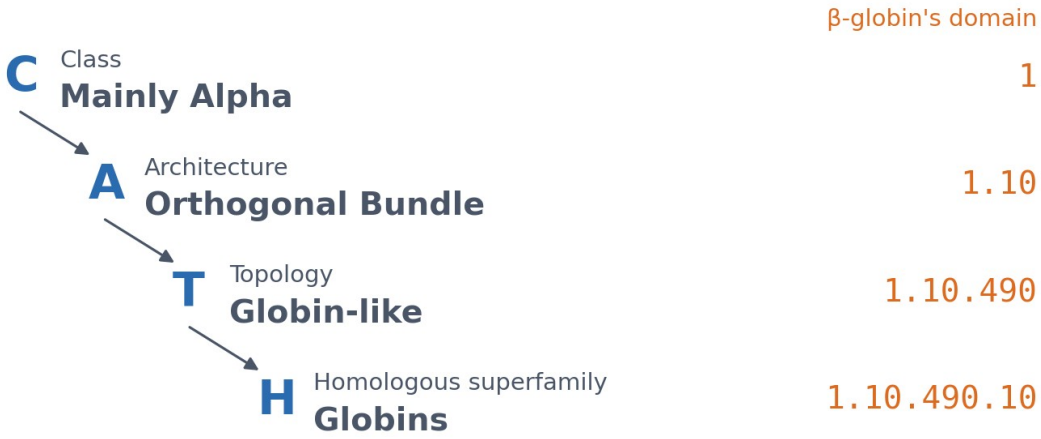
Break

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BLOCK 3 · ABOUT 30 MIN

Domains, states, and structure as a prediction target

CATH classifies domains in four levels

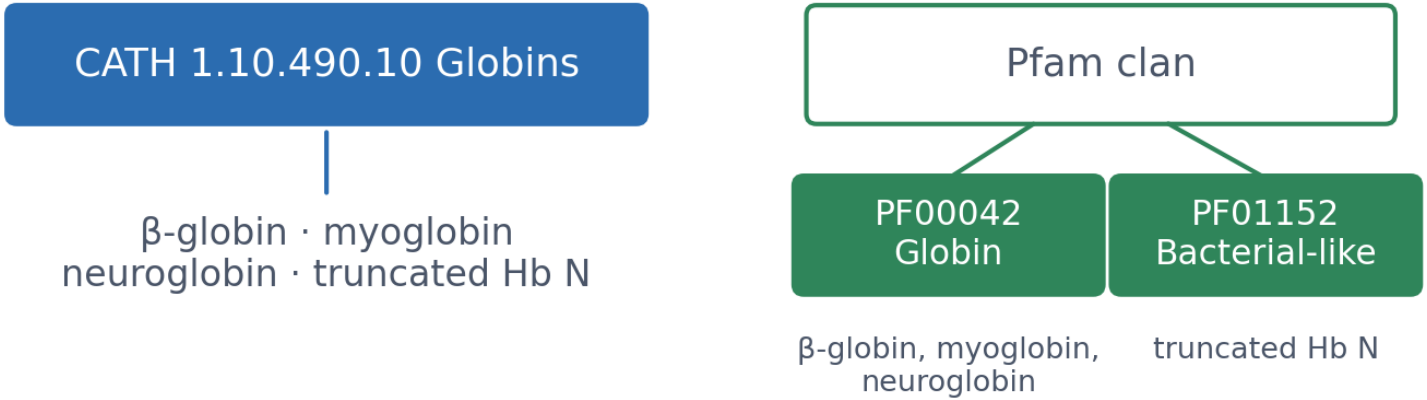


- **Domain: a compact, reusable unit**
- Grouped by 3D shape
- Myoglobin, neuroglobin, truncated Hb: same code

Orengo et al. (1997); live PDBe CATH lookups. GFP: class 2 (Mainly Beta), Beta Barrel.

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Structure sees one superfamily, sequence sees two families



CATH puts the truncated haemoglobin in the same superfamily as β-globin. Pfam, working from sequence, needs a separate family (PF01152, 'Bacterial-like globin') and joins it to PF00042 'Globin' only in a clan.

Live PDBe CATH and InterPro lookups (course.bioinfo.se/session06.html).

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Other classifications, other philosophies

SCOP

1995

- Expert manual curation

ECOD

2014

- Organised by evolutionary relationship

TED

2024

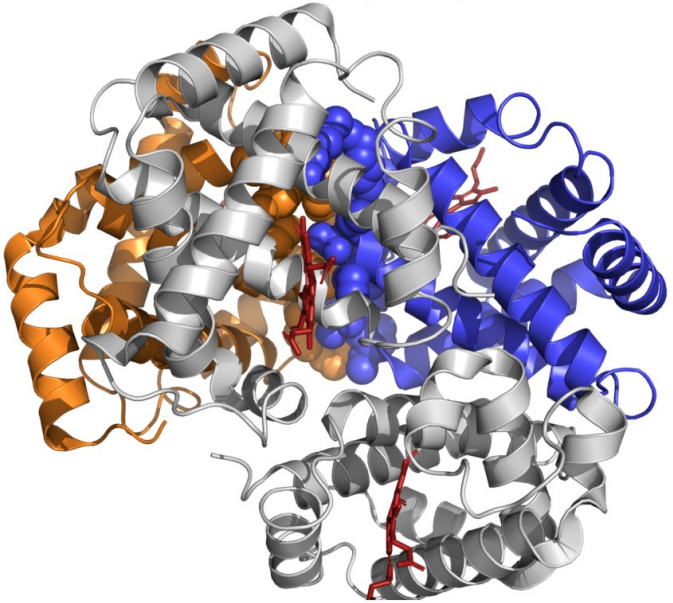
- Parsed automatically from the whole AlphaFold Database

Murzin et al. (1995); Cheng et al. (2014); Lau, Bordin et al. (2024).

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Interfaces and ligands: where function happens

Deoxyhaemoglobin: interface atoms of one α chain (blue) and the β chain it touches (orange) as spheres; haem in red



- **Interface: where two chains touch**
- More hydrophobic, more conserved
- Ligands: haem, ions, small molecules
- Predicting them: Session 12

Deoxyhaemoglobin (2HHB): atoms within 5 Å of the other chain as spheres, rendered with PyMOL.

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Docking: three routes to a complex

Shape and physics

1992

- FFT scores every relative translation at once
- GRAMM, ZDOCK

Data

2003

- HADDOCK: mutations and NMR mark the interface

Templates

2012

- Build on a related complex of known structure

Katchalski-Katzir et al. (1992); Vakser (1995); Chen, Li & Weng (2003); Dominguez, Boelens & Bonvin (2003); Kundrotas, Zhu, Janin & Vakser (2012).

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Judging a complex: each chain right, the pair wrong

- **Ligand RMSD: superimpose the receptor, measure the other chain**
- Interface RMSD: only over the residues in contact
- Or count contacts: what fraction of the real ones are reproduced
- **DockQ combines them into one number, 0 to 1**

Basu & Wallner (2016), PLoS One 11:e0161879.

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CAPRI: straightforward with a template, hard without

- **CAPRI since 2001: docking's CASP**
- With a template: 'very good to excellent'
- **Without one: medium or high quality for only 3 of 11 hard targets**
- What changed that: Session 12

3 / 11
hard targets modelled well, CASP13 (2018)

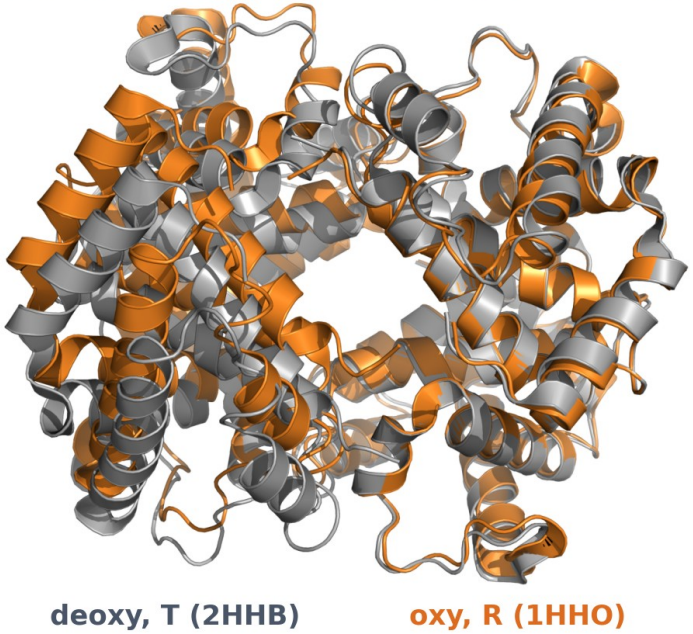
2001
first CAPRI assessment

Janin et al. (2003); Lensink et al. (2019), Proteins 87:1200.

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One protein, two structures: haemoglobin T and R

Superimposed on one α/β pair: the other pair has rotated



- **Each chain hardly changes (TM 0.98)**
- The α/β pairs rotate ≈13-15°
- β-β Fe-Fe: 39.5 → 34.6 Å
- Cooperative oxygen binding

Baldwin & Chothia (1979); notebook (Fe-Fe distances); 1HHO biological assembly.

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A PDB entry is one snapshot, not 'the' structure

- **States depend on ligands, partners, conditions**
- Matters whenever a model is compared with 'the' experiment
- **AlphaFold usually predicts one state (Session 12)**

4.9 Å

β-β haem Fe-Fe contraction, T → R

0.98

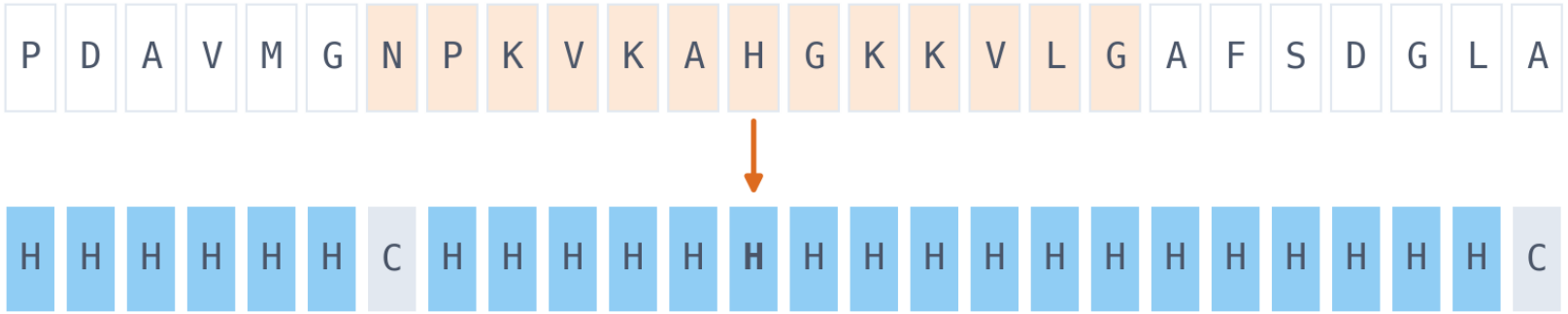
TM-score of each chain, T vs R

Notebook: session06-protein-structure-in-code.ipynb.

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Secondary structure as a prediction target

input: a window of sequence (or profile) around each residue



target: H, E or C for every residue, from solved structures (DSSP); β-globin residues 51-76

Input: the sequence (or its profile), one window per residue. Target: H, E or C for every residue, with labels from solved structures. What could help? The residue itself, its neighbours, and how the position varies among homologs (Session 5).

β-globin residues 51-76 with the PDB entry's secondary structure; window 57-69 around His63 (as in Session 7).

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Four ways to predict a whole fold

physics	comparative	contact prediction	deep learning
lowest-energy chain; Rosetta fragments	a related known structure (S4-5)	coevolving pairs in alignments (S5)	AlphaFold2, CASP14 (S12)

CASP: blind tests on unpublished structures since 1994

Physics is rigorous but expensive; comparative modelling needs a template; coevolution fills in a contact map; deep learning (AlphaFold2, 2020) combined all of it. 2024 Nobel Prize in Chemistry: Baker, Hassabis, Jumper.

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Residues in contact mutate together

A	K	L	E	D	G	E
A	R	L	E	E	G	D
S	K	V	E	D	A	E
A	E	L	Q	K	G	K
S	E	V	Q	K	A	R

→ these two residues are probably in contact

illustrative: K/R in one column goes with E/D in the other, and vice versa

- **Large alignments reveal coevolving pairs**
- Coevolving pairs → a contact map
- Session 12: AlphaFold

Illustrative alignment, not real sequences. See course.bioinfo.se/session06.html, 'Predicting the whole fold'.

One structure becomes many

ensembles	inside the cell	Nobel Prize 2024
AF-Cluster: AlphaFold2 samples both folds of the fold-switching protein KaiB	cryo-electron tomography: RNA polymerase and ribosome coupled, inside Mycoplasma cells	Baker: computational design Hassabis & Jumper: structure prediction
Wayment-Steele et al. 2024	*O'Reilly et al. 2020*	*Chemistry*

haemoglobin has (at least) two: T and R. Can one predicted structure ever be enough?

- Proteins move: ensembles, not snapshots
- Structures inside cells, by cryo-ET
- Nobel Prize in Chemistry 2024
- **Can one predicted structure ever be enough?**

Wayment-Steele et al. (2024), Nature 625:832; O'Reilly et al. (2020), Science 369:554; nobelprize.org (2024).

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The course's path from sequence to structure



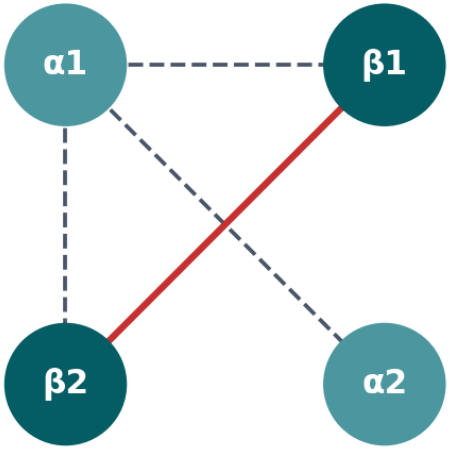
Today: structures as data, compared by alignment, TM-score, distance maps and Foldseek, and the reminder that one entry is one state. Next: models that learn the steps from sequence to structure.

See course.bioinfo.se/session06.html, 'Predicting the whole fold'.

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Which iron–iron distance changes the least?

T (2HHB) to R (1HHO): each chain hardly changes shape, but the β1–β2 iron–iron distance shrinks by about 5 Å.



Fe–Fe distances, T (2HHB) → R (1HHO)

β1–β2: about 5 Å shorter

each chain: Cα RMSD below 1 Å

Which pair changes the least?

A α1–β1, the "fixed" interface

B α1–β2, the "sliding" interface

C α1–α2

D None: all shrink by about 5 Å

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

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Which iron–iron distance changes the least?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Which iron–iron distance changes the least?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-3.ipynb

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Which iron–iron distance changes the least?

Answer again

Same question, after the discussion.



quiz.bioinfo.se/live

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Which iron–iron distance changes the least?

Results, and who explains

Both rounds side by side.

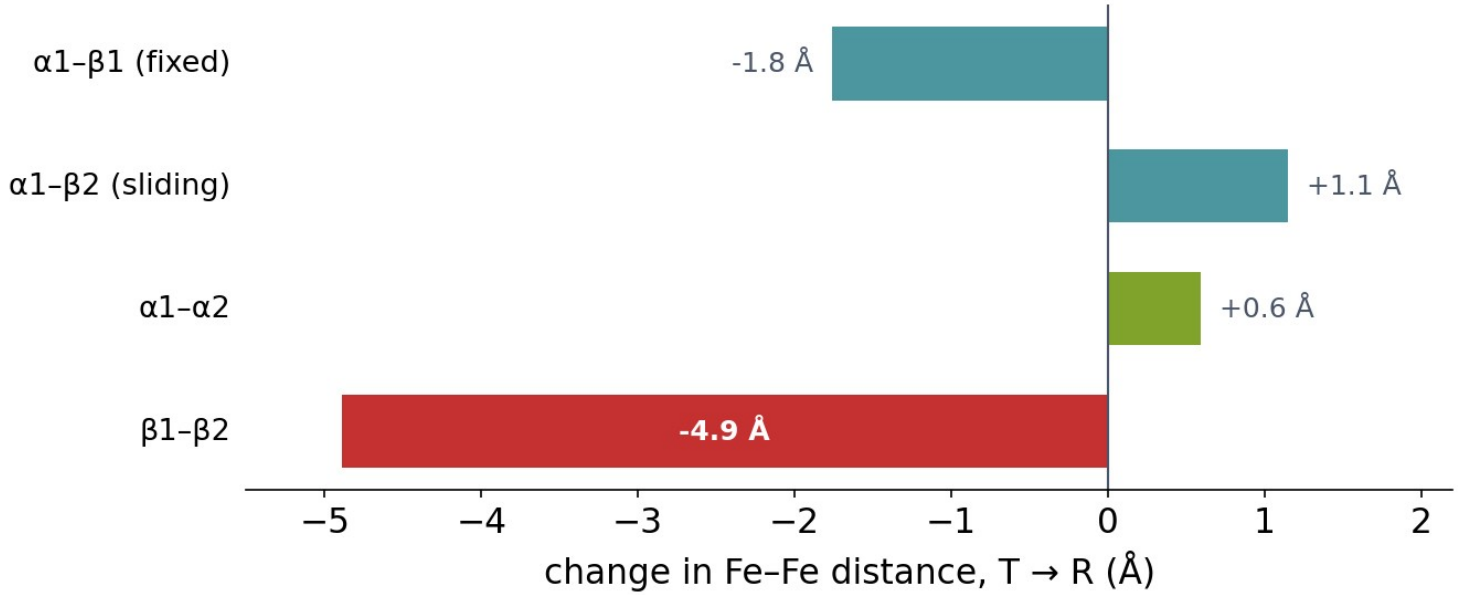
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 3 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session06-discussion-3.ipynb

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The answer: α1–α2 barely moves; one distance is not the change



the two α/β pairs rotate about 13° as rigid bodies

From T to R, β1–β2 shrinks by 4.9 Å but α1–α2 changes by only 0.6 Å. The chains hardly change shape: the two α/β pairs rotate about 13° against each other as rigid bodies, and a rotation moves each atom by an amount that depends on its distance from the axis. Describing a conformational change needs the rigid units and their motion, not one distance.

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In up to three words: what was the hardest part of today?

Four levels

sequence to 3D, contact maps

Comparing in 3D

RMSD, TM-score, DALI, Foldseek

Domains and states

CATH, T and R, prediction

Answer now

Scan the code, or go to
quiz.bioinfo.se/live

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The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

Session 7 next

Session 7: the PSSM's weighted sum becomes a perceptron with learned weights
First target: the secondary structure you saw today